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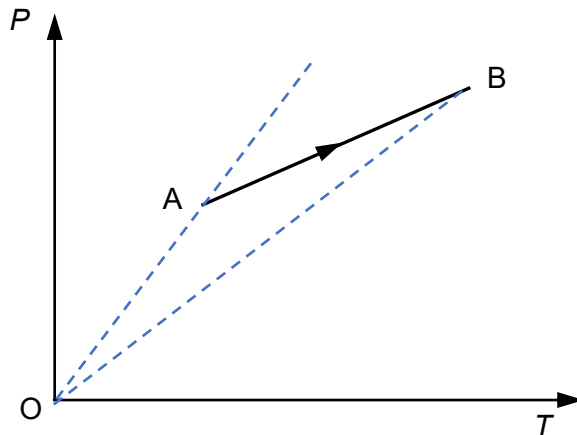
On the P - T diagram, using $p = \left(\frac{nR}{V}\right)T$,
the volume is inversely proportional to the
gradient of line passing through the origin.

gradient of OA > gradient of OB

Thus, volume of the gas increases, and
positive work is done by the gas in the
process AB (or work done on the gas is
negative, $W < 0$).

Temperature increases from A to B implies
that internal energy of the ideal gas
increases in the process, i.e. $\Delta U > 0$.

By First Law of Thermodynamics, $\Delta U = W + Q \Rightarrow Q = (\Delta U - W) > 0$
Thus heat is supplied to the gas.



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The pendulum performs simple harmonic motion at small angles.